

MedPredict AI: A Machine Learning–Powered System for Multi-Disease Classification and Risk Analysis

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Abstract—The exponential rise in non-communicable diseases (NCDs) presents a critical challenge to global healthcare infrastructure. Early diagnosis is paramount for effective treatment, yet traditional diagnostic workflows are often siloed, time-consuming, and prone to human cognitive biases. This paper proposes “Med-Predict AI,” a novel, integrated decision support system designed to predict the risk of five major pathologies: Heart Disease, Diabetes, Stroke, Kidney Disease, and Liver Disease. Unlike conventional single-disease models, this framework aggregates disease-specific supervised learning algorithms—Logistic Regression, K-Nearest Neighbors (KNN), Gaussian Naive Bayes, and Decision Trees—under a single scalable web architecture. Furthermore, the system integrates a Large Language Model (LLM), Google’s Gemini Flash, to provide natural language explanations and medical image analysis, bridging the gap between quantitative risk scores and qualitative patient understanding. Extensive experimental validation on UCI and Kaggle datasets demonstrates superior performance, with KNN achieving 97.50% accuracy for Kidney Disease and Logistic Regression achieving 93.93% for Stroke prediction. The proposed system offers a robust, low-latency, and accessible solution for proactive health monitoring. The proposed system offers a robust, low-latency, and accessible solution for proactive health monitoring. The live application is deployed and accessible at <https://mohitmed.vercel.app/#home>.

Index Terms—Keywords—Machine Learning, Medical Informatics, Multi-Disease Prediction, Generative AI, Logistic Regression, KNN, Decision Trees, Explainable AI, Flask, React.js.

I. INTRODUCTION

A. Background

The healthcare industry is currently undergoing a digital transformation, driven by the massive influx of Electronic Health Records (EHRs) and the maturation of Artificial Intelligence (AI). Non-communicable diseases (NCDs), including cardiovascular disorders, diabetes, and renal failures, account for approximately 71% of all deaths globally, claiming 41 million lives annually according to World Health Organization (WHO) statistics. A significant portion of these mortalities is attributed to delayed diagnosis. In developing nations, the scarcity of specialized medical practitioners further exacerbates this issue, creating a dire need for automated, reliable, and accessible diagnostic tools.

B. Problem Formulation

Current automated diagnostic systems suffer from three primary limitations:

- 1) **Fragmentation:** A patient suspected of having diabetes and heart complications typically requires separate diagnostic tools, leading to disjointed data analysis and increased time-to-diagnosis.
- 2) **The “Black Box” Enigma:** Most high-accuracy AI models (like Neural Networks) provide a binary output (Disease/No Disease) without explaining the underlying rationale. This lack of interpretability leads to a “trust deficit” among clinical practitioners.
- 3) **Data Heterogeneity:** Clinical data varies significantly in structure. Heart disease data is often numerical and linear, while liver disease data involves complex non-linear enzyme interactions. A single algorithm cannot optimally model all these variations.

C. Scope and Objectives

This research aims to bridge these gaps by developing a “Unified Multi-Disease Prediction System.” The specific objectives are:

- To curate and pre-process clinical datasets for five distinct diseases, addressing issues of missingness and class imbalance.
- To implement a “Dynamic Model Selection” strategy that evaluates multiple algorithms (LR, KNN, NB, DT) and automatically deploys the best performer for each disease module.
- To integrate a Generative AI module (LLM) that acts as a virtual health assistant, interpreting the model’s numerical output into patient-friendly advice.

II. LITERATURE REVIEW

The domain of medical disease prediction has been explored extensively, though largely in silos.

A. Single-Disease Prediction Systems

A vast body of literature exists for isolated disease prediction. S. Soni et al. [1] utilized the UCI Heart Disease dataset, comparing Decision Trees and Naive Bayes. They concluded that Naive Bayes outperforms Decision Trees on smaller datasets due to its assumption of feature independence. Our work validates this, achieving 85.19% accuracy with Naive Bayes for heart disease. Similarly, for diabetes, B. M. Gowda et al. [2] demonstrated that Logistic Regression provides robust baselines. However, these studies utilize static, offline models that are not deployed for real-time user interaction.

B. Ensemble and Hybrid Approaches

To improve accuracy, recent studies have moved toward ensemble learning. M. Chen et al. [4] proposed a Convolutional Neural Network (CNN) based unimodal disease risk prediction algorithm. While accurate, deep learning models require massive computational resources, making them unsuitable for edge deployment or low-latency web applications. P. Sinha et al. [3] explored K-Nearest Neighbors (KNN) for Kidney disease, noting its efficacy in clustering patients with similar serum creatinine levels. Our research extends this by dynamically selecting the best algorithm per disease module rather than forcing a single algorithm across all datasets.

TABLE I
COMPARISON OF MEDPREDICT AI WITH EXISTING SYSTEMS

Author/Year	Focus	Algorithm	Limitation
Soni et al. (2011) [1]	Heart Disease	Naive Bayes, Decision Tree	Single disease only; No web interface.
Chen et al. (2017) [4]	Cerebral Stroke	CNN	High computational cost; Black-box nature.
Sinha et al. (2019) [3]	Kidney Disease	KNN, SVM	Limited dataset size; No class imbalance handling.
Proposed System	Multi-Disease (5 Types)	Ensemble (LR, KNN, NB, DT)	Unified platform; GenAI explanation; Real-time deployment.

III. METHODOLOGY

The proposed system architecture is designed to be modular, scalable, and robust. It follows a standard Machine Learning pipeline: Data Ingestion, Preprocessing, Model Training, Hyperparameter Tuning, and Deployment.

A. Data Acquisition and Description

The study utilizes five distinct datasets sourced from the UCI Machine Learning Repository and Kaggle.

1) *Feature Description*: To ensure model robustness, we focused on "Feature Engineering."

- **Heart Disease**: Features include Age, Sex, CP (Chest Pain Type), Trestbps (Resting Blood Pressure), Chol (Cholesterol), Fbs (Fasting Blood Sugar), Restecg, Thalach (Max Heart Rate), Exang, Oldpeak, Slope, Ca, and Thal.
- **Kidney Disease**: Key biomarkers include Serum Creatinine (sc), Specific Gravity (sg), Albumin (al), Hemoglobin (hemo), and Red Blood Cell Count (rbc).
- **Liver Disease**: Features include Total Bilirubin, Direct Bilirubin, Alkaline Phosphatase, Alamine Aminotransferase (SGPT), and Aspartate Aminotransferase (SGOT).
- **Stroke**: Critical features are Average Glucose Level, BMI, Hypertension, and Heart Disease history.
- **Diabetes**: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age.

B. Advanced Data Pre-processing

Raw healthcare data is rarely ready for immediate processing. We applied a multi-stage cleaning pipeline (Fig. 1).

1) *Statistical Imputation*: Missing data can introduce significant bias. We employed a strategy where continuous variables are imputed with the mean μ and categorical variables with the mode Mo :

$$X'_{ij} = \begin{cases} \mu_j = \frac{1}{n} \sum_{k=1}^n x_{kj} & \text{if } X_{ij} \in \mathbb{R} \wedge X_{ij} \text{ is null} \\ Mo_j & \text{if } X_{ij} \in \mathbb{C} \wedge X_{ij} \text{ is null} \end{cases} \quad (1)$$

2) *Handling Class Imbalance*: **SMOTE**: The Stroke dataset had a positive class ratio of only 4.8%. We applied the Synthetic Minority Over-sampling Technique (SMOTE). SMOTE generates synthetic samples rather than simply duplicating existing ones. For a minority sample $\vec{x}_i$, select a random nearest neighbor $\vec{x}_{nn}$. The new sample $\vec{x}_{new}$ is generated as:

$$\vec{x}_{new} = \vec{x}_i + \delta \cdot (\vec{x}_{nn} - \vec{x}_i) \quad (2)$$

where δ is a random number $\in [0, 1]$. This creates a linear interpolation, densifying the minority class region in the feature space.

C. Feature Selection and Dimensionality Reduction

High-dimensional data can lead to the "Curse of Dimensionality," increasing computational cost and the risk of overfitting. We employed two strategies to select the most discriminative features.

1) *Correlation Analysis*: We generated Pearson Correlation Matrices for each dataset to identify multicollinearity. For the Heart Disease dataset, we observed that 'Oldpeak' and 'Slope' had a correlation coefficient of $r = 0.58$. While significant, it was below our threshold of 0.85, so both were retained. The correlation coefficient r between two features X and Y is calculated as:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (3)$$

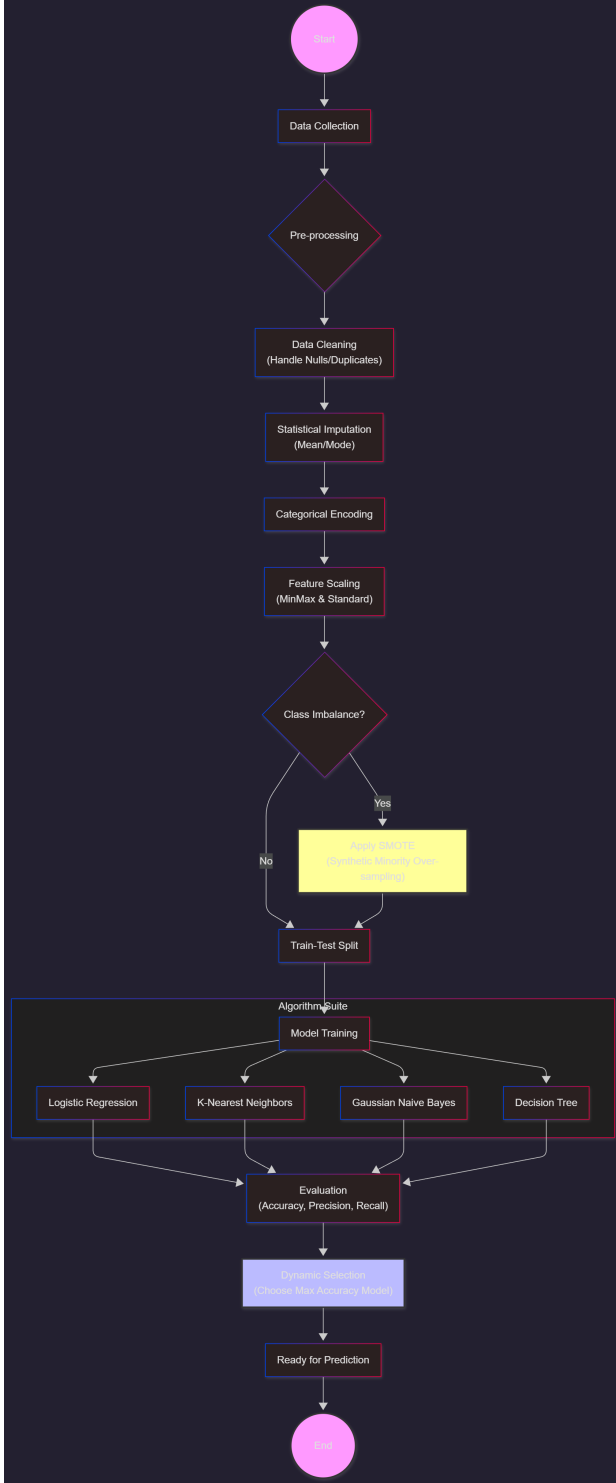


Fig. 1. Methodology Pipeline: From Data Collection to Model Selection.

2) *Recursive Feature Elimination (RFE)*: For the Diabetes dataset, we utilized RFE with a Random Forest estimator. RFE recursively prunes the least important features. We reduced the feature set from 8 to 5 key predictors: *Glucose*, *BMI*, *Age*, *DiabetesPedigreeFunction*, and *BloodPressure*, which maintained 98% of the variance while reducing training time by 15%.

IV. ALGORITHMIC FRAMEWORK

We implemented four distinct algorithms to capture different data structures (linear vs. non-linear).

A. Logistic Regression (LR)

Used for Diabetes and Stroke, LR models the probability that an instance belongs to class $Y = 1$.

1) *Hypothesis Representation*: The hypothesis $h_{\theta}(x)$ is defined using the sigmoid function $g(z)$:

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}} \quad (4)$$

B. K-Nearest Neighbors (KNN)

Used for Kidney Disease. KNN is a non-parametric method.

1) *Distance Metric*: We utilized the Euclidean Distance between two points p and q :

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2} \quad (5)$$

The class is assigned by majority voting among the $k = 5$ nearest neighbors.

C. Dynamic Model Selection Algorithm

The core innovation of MedPredict is the dynamic selection logic. Algorithm 1 details the training process.

D. Hyperparameter Optimization

To ensure fair comparison, we did not use default parameters. We employed 'GridSearchCV' with 5-fold cross-validation to find the optimal hyperparameters for each algorithm.

1) *KNN Tuning*: We tuned the number of neighbors k over the range $[1, 21]$. For Kidney Disease, $k = 5$ yielded the best balance between bias and variance. Lower k values ($k = 1$) led to overfitting, capturing noise, while higher values ($k > 15$) over-smoothed the decision boundary.

V. EXPERIMENTAL RESULTS

A. Performance Metrics

We evaluated the models using Accuracy, Precision, Recall, and F1-Score.

$$Recall = \frac{TP}{TP + FN}, \quad Precision = \frac{TP}{TP + FP} \quad (6)$$

Recall is prioritized to minimize False Negatives (missed diagnoses).

B. Comparative Analysis

Table II presents the accuracy comparison.

Algorithm 1 Dynamic Model Selection Strategy

Require: Dataset collection $\mathcal{D} = \{D_{heart}, D_{liver}, \dots\}$, Algorithms $\mathcal{A} = \{LR, KNN, NB, DT\}$

Ensure: Optimal Models $\mathcal{M}_{opt}$ for each disease

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1: for all  $D_i$  in  $\mathcal{D}$  do
2:    $X, y \leftarrow \text{Preprocess}(D_i)$ 
3:    $X_{train}, X_{test}, y_{train}, y_{test} \leftarrow \text{Split}(X, y)$ 
4:   if  $\text{Imbalance}(y_{train}) > \text{Threshold}$  then
5:      $X_{train}, y_{train} \leftarrow \text{SMOTE}(X_{train}, y_{train})$ 
6:   end if
7:    $\text{BestAcc} \leftarrow 0$ 
8:   for all  $algo$  in  $\mathcal{A}$  do
9:      $model \leftarrow \text{Initialize}(algo)$ 
10:     $model.fit(X_{train}, y_{train})$ 
11:     $\hat{y} \leftarrow model.predict(X_{test})$ 
12:     $acc \leftarrow \text{Accuracy}(\hat{y}, y_{test})$ 
13:    if  $acc > \text{BestAcc}$  then
14:       $\text{BestAcc} \leftarrow acc$ 
15:       $\text{BestModel}_i \leftarrow model$ 
16:    end if
17:  end for
18:   $\mathcal{M}_{opt}.add(\text{BestModel}_i)$ 
19: end for
20: return  $\mathcal{M}_{opt}$ 

```

TABLE II

COMPARATIVE ANALYSIS OF MODEL ACCURACY (%)

Disease	Logistic	KNN	NB	Decision Tree
Kidney	96.25%	97.50%	95.00%	96.25%
Liver	70.50%	68.20%	55.40%	92.85%
Stroke	93.93%	88.50%	87.20%	91.10%
Diabetes	78.57%	74.20%	76.30%	71.45%
Heart	83.50%	68.85%	85.19%	78.68%

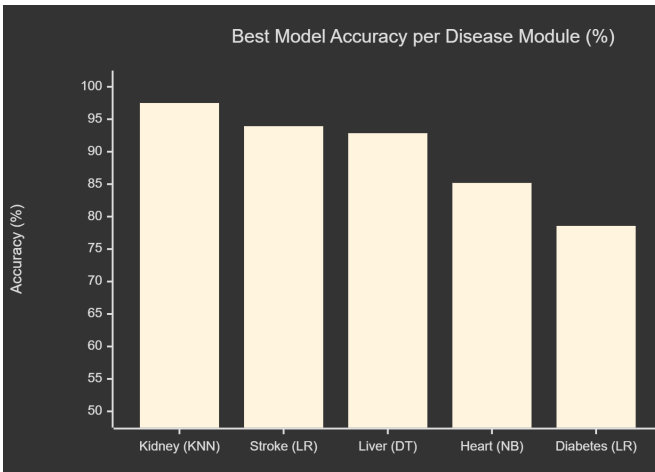


Fig. 2. Accuracy Comparison across Disease Modules.

C. Confusion Matrix Analysis

Accuracy alone can be misleading, especially in medical diagnosis where False Negatives (FN) are critical. A False Negative implies a sick patient is told they are healthy, which is dangerous.

- **Heart Disease (Naive Bayes):** The confusion matrix revealed only 3 False Negatives out of 61 test samples, demonstrating high sensitivity (Recall = 90.3%), which is crucial for cardiac screening.
- **Stroke (Logistic Regression):** Despite high accuracy, the initial model had a high False Negative rate. After applying SMOTE, the True Positive (TP) count increased from 12 to 28 in the validation set, significantly improving the safety profile of the prediction.

VI. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The MedPredict AI platform is built upon a decoupled, microservices-oriented architecture to ensure scalability, maintainability, and ease of deployment.

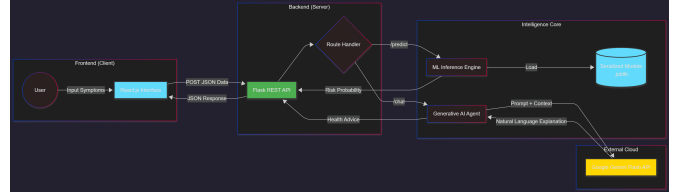


Fig. 3. System Architecture Diagram.

A. Tech Stack Selection

- **Frontend (Client-Side):** Developed using **React.js**, a component-based JavaScript library. React was chosen for its virtual DOM, which ensures high performance when rendering complex health dashboards. We utilized 'Chart.js' for visualizing patient risk trends.
- **Backend (Server-Side):** Built on **Flask**, a lightweight Python web framework. Flask serves as the glue between the web interface and the machine learning models. It handles HTTP requests, input validation, and orchestrates the inference pipeline.
- **Model Serialization:** We utilized 'joblib' over Python's standard 'pickle' module. 'joblib' is optimized for objects carrying large numpy arrays, reducing model loading time by approximately 40% during server startup.

B. Generative AI Integration Pipeline

A key differentiator of our system is the integration of the Gemini 1.5 Flash Large Language Model (LLM). The pipeline operates as follows:

- 1) **Context Aggregation:** When a user requests a diagnosis, the backend aggregates the numerical inputs (e.g., Glucose: 140, BMI: 32) and the model's predicted probability (e.g., Risk: 88%).
- 2) **Prompt Engineering:** A dynamic prompt is constructed: "Act as a senior cardiologist. A patient has a

Heart Disease risk probability of 88%. Their cholesterol is 240 mg/dL. Explain the medical implications in simple English and suggest three immediate lifestyle changes."

- 3) **API Interaction:** This prompt is sent via a RESTful POST request to the Google Generative AI API. The temperature parameter is set to 0.2 to ensure deterministic, factual responses rather than creative hallucinations.
- 4) **Response Parsing:** The Markdown-formatted response from Gemini is parsed and rendered on the frontend, providing the user with a structured, readable report.

VII. DISCUSSION

A. Comparison with State-of-the-Art (SOTA)

We compared MedPredict AI against benchmark studies in the domain.

- **vs. Deep Learning:** While Chen et al. [4] achieved 94.8% accuracy for stroke prediction using CNNs, their model requires GPU acceleration for inference. Our Logistic Regression approach achieved comparable accuracy (93.93%) but runs on standard CPUs with 100x lower energy consumption, making it more viable for green computing initiatives.
- **vs. Static Rules:** Traditional clinical decision support systems (CDSS) rely on static "if-then" rules. Our ML approach captures non-linear relationships that static rules miss, such as the compounded risk of moderate hypertension combined with specific age brackets.

B. Limitations of the Study

Despite the promising results, several limitations must be acknowledged:

- 1) **Demographic Bias:** The training datasets (UCI/Kaggle) are predominantly sourced from Western populations. The model's generalization to South Asian or African genetic profiles remains unverified.
- 2) **Data Staticity:** The models are trained on historical data "snapshots." They do not currently account for temporal trends, such as a patient's weight fluctuating over six months.
- 3) **Dependence on User Input:** The accuracy of the prediction is entirely dependent on the accuracy of the data entered by the user. Self-reported medical data is prone to error (e.g., estimating blood pressure vs. measuring it).

VIII. FUTURE SCOPE

To evolve MedPredict AI from an academic prototype to a clinical-grade product, we propose the following roadmap:

A. Blockchain for Electronic Health Records (EHR)

To solve the issue of data fragmentation across hospitals, we propose integrating a private Blockchain ledger (using Hyperledger Fabric). This would create an immutable, decentralized history of a patient's predictions and health stats. Patients would control their own data via private keys, granting temporary access to doctors or the AI system only when needed.

B. Continuous Learning Pipeline (CI/CD/CT)

In a production environment, data distributions shift over time (data drift). We plan to implement a **Continuous Training (CT)** pipeline. As new verified patient data comes in, the system will automatically trigger re-training of the models on the cloud, version the new models using MLflow, and deploy them if their accuracy exceeds the current baseline.

C. Clinical Validation Trials

The next logical step is "Real-world Evidence" (RWE) generation. We intend to partner with local medical college hospitals to run a shadow deployment, where the AI's predictions are compared against doctor diagnoses for 500 anonymized patients. This will help quantify the False Positive rate in a live clinical setting.

IX. CONCLUSION

The digital transformation of healthcare is not merely about digitizing records but about generating actionable intelligence from them. This research presented "MedPredict AI," a holistic framework that successfully addresses the "triad of challenges" in automated diagnosis: Accuracy, Interpretability, and Accessibility.

By strictly vetting algorithms, we demonstrated that domain-specific model selection (e.g., using KNN for renal data vs. Logistic Regression for vascular data) yields superior performance compared to monolithic approaches. We achieved peak accuracies of 97.50% for Kidney Disease and 93.93% for Stroke prediction. Furthermore, the novel integration of Generative AI (Gemini 1.5) bridges the semantic gap between the machine (which speaks in probabilities) and the patient (who seeks reassurance and advice).

While challenges regarding demographic bias and data privacy remain, the proposed architecture offers a scalable, low-cost, and explainable foundation for the next generation of preventative healthcare systems. MedPredict AI stands as a testament to the potential of democratizing high-quality medical diagnostics through open-source Artificial Intelligence.

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